

Read a file stored in HDFS using a Java program (URLCat.java) that uses Hadoop's Java API.

REQUIREMENTS CHECKLIST

Make sure the following are ready:

-  Hadoop is installed and working
 -  Java is installed (javac, java commands work)
 -  HDFS is running (start-dfs.sh)
 -  `hdfs dfs -ls /` works without error
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STEP-BY-STEP EXECUTION

◆ Step 1: Create a Sample Input File Locally

📍 **Where:** Terminal

```
echo "Hello, this is a sample input file for HDFS read test." > inputfile.txt
```

Verify the file:

```
cat inputfile.txt
```

◆ Step 2: Upload the File to HDFS

📍 **Where:** Terminal

```
hdfs dfs -copyFromLocal inputfile.txt /
```

Verify upload:

```
hdfs dfs -ls /
```

You should see `/inputfile.txt` listed.

◆ Step 3: Write the Java Code (URLCat.java)

📍 **Where:** Use gedit, nano, or any text editor in terminal

```
gedit URLCat.java
```

Paste the following:

```
import java.io.InputStream;
```

```
import java.net.URL;

import org.apache.hadoop.fs.FsUrlStreamHandlerFactory;

import org.apache.hadoop.io.IOUtils;

public class URLCat {

    static {

        URL.setURLStreamHandlerFactory(new FsUrlStreamHandlerFactory());

    }

    public static void main(String[] args) throws Exception {

        InputStream in = null;

        try {

            in = new URL(args[0]).openStream(); // HDFS file URL

            IOUtils.copyBytes(in, System.out, 4096, false);

        } finally {

            IOUtils.closeStream(in);

        }

    }

}
```

Save and close.

◆ Step 4: Compile the Java Program

📌 **Where:** Terminal

```
javac -classpath `hadoop classpath` URLCat.java
```

If there is no error, it will generate a URLCat.class file.

◆ Step 5: Run the Java Program

📌 **Where:** Terminal

```
java -classpath `hadoop classpath` URLCat hdfs://localhost:9000/inputfile.txt
```

✓ You should now see:

Hello, this is a sample input file for HDFS read test.

Troubleshooting Tips

Problem	Solution
FileNotFoundException	Make sure the file exists in HDFS using <code>hdfs dfs -ls /</code>
NativeCodeLoader warning	Safe to ignore
setURLStreamHandlerFactory() error	Only call once per JVM; don't run multiple URLCat.java in the same JVM session

Summary of What You Did

Step Action

- 1 Created a sample file locally
 - 2 Uploaded file to HDFS using `-copyFromLocal`
 - 3 Wrote Java code using HDFS Java API
 - 4 Compiled Java file with Hadoop classpath
 - 5 Ran the program with an HDFS file URL
-  Successfully read and printed file content
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Would you like to now perform the **write operation to HDFS using Java** as well?